

B1 Fluids, 2021 — Model Solutions

Question 1 — Stokes flow between hinged plates

Problem. High vs low Re; biharmonic streamfunction; viscous flow between two long, wide plates hinged at $r = 0$, opening at constant $\dot{\alpha}$; verify the quoted ψ , find breakdown scales, flux, force scaling, and discuss the scallop theorem.

(a) High vs low Reynolds number

$\text{Re} = UL/\nu$ measures inertia/viscous force ratio. High Re ($\gg 1$): inertia dominates; viscosity confined to thin boundary layers $\delta \sim L/\sqrt{\text{Re}}$; turbulence; form drag. Low Re ($\ll 1$): linear Stokes equation $\nabla p = \mu \nabla^2 \mathbf{v}$; time-reversible, instantaneous, drag $\propto U$.

Swimming human: $L \sim 1 \text{ m}$, $U \sim 1 \text{ m s}^{-1}$, $\nu = 10^{-6} \text{ m}^2 \text{ s}^{-1}$. Plug in numbers:

$$\text{Re} \sim \frac{(1)(1)}{10^{-6}}.$$

Evaluate:

$$\text{Re} \sim 10^6.$$

$\text{Re} \sim 10^6 \gg 1$: high-Re regime.

(b) Streamfunction and biharmonic equation

Polar incompressibility:

$$\nabla \cdot \mathbf{v} = \frac{1}{r} \frac{\partial(rv_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta}.$$

Substitute $v_r = (1/r)\partial_\theta \psi$, $v_\theta = -\partial_r \psi$:

$$\nabla \cdot \mathbf{v} = \frac{1}{r} \frac{\partial}{\partial r} \left(\frac{\partial \psi}{\partial \theta} \right) - \frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{\partial \psi}{\partial r} \right).$$

Equality of mixed partials:

$$\nabla \cdot \mathbf{v} = 0.$$

Vorticity in 2D: $\boldsymbol{\omega} = \omega \hat{\mathbf{z}}$, $\omega = -\nabla^2 \psi$. At low Re, drop inertia in NS:

$$\nabla p = \mu \nabla^2 \mathbf{v}. \quad (1)$$

Take curl of (??):

$$\nabla \times \nabla p = 0 = \mu \nabla^2 (\nabla \times \mathbf{v}).$$

Use $\nabla \times \mathbf{v} = \omega \hat{\mathbf{z}}$ in 2D:

$$0 = \mu \nabla^2 \omega.$$

Substitute $\omega = -\nabla^2 \psi$:

$\nabla^{22} \psi = \nabla^4 \psi = 0.$

(2)

(c) Boundary conditions and verification

BCs. No-slip on the fixed plate ($\theta = 0$):

$$v_r(r, 0) = 0, \quad v_\theta(r, 0) = 0 \implies \left. \frac{\partial \psi}{\partial \theta} \right|_0 = 0, \quad \left. \frac{\partial \psi}{\partial r} \right|_0 = 0.$$

Rigid rotation of upper plate ($\theta = \alpha$): $\mathbf{v}_{\text{plate}} = r\dot{\alpha}\hat{\boldsymbol{\theta}}$:

$$v_r(r, \alpha) = 0, \quad v_\theta(r, \alpha) = r\dot{\alpha} \implies \left. \frac{\partial \psi}{\partial \theta} \right|_\alpha = 0, \quad \left. \frac{\partial \psi}{\partial r} \right|_\alpha = -r\dot{\alpha}.$$

Ansatz. BC linear in r forces $n = 2$:

$$\psi(r, \theta) = r^2 f(\theta), \quad f(\theta) = A \cos 2\theta + B \sin 2\theta + C\theta \cos 2\theta + D\theta \sin 2\theta.$$

Reduce BCs by dividing r^2 and $2r$:

$$f(0) = 0, \quad f'(0) = 0, \quad f(\alpha) = -\frac{1}{2}\dot{\alpha}, \quad f'(\alpha) = 0.$$

Apply $f(0) = 0$:

$$A = 0.$$

Differentiate f :

$$f'(\theta) = -2A \sin 2\theta + 2B \cos 2\theta + C(\cos 2\theta - 2\theta \sin 2\theta) + D(\sin 2\theta + 2\theta \cos 2\theta).$$

Apply $f'(0) = 0$:

$$2B + C = 0 \implies C = -2B.$$

Reduced f :

$$f(\theta) = B \sin 2\theta - 2B\theta \cos 2\theta + D\theta \sin 2\theta.$$

Reduced f' :

$$f'(\theta) = 4B\theta \sin 2\theta + D(\sin 2\theta + 2\theta \cos 2\theta).$$

Write $s = \sin 2\alpha$, $c = \cos 2\alpha$. The two upper-plate conditions become

$$4B\alpha s + D(s + 2\alpha c) = 0,$$

$$B(s - 2\alpha c) + D\alpha s = -\frac{1}{2}\dot{\alpha}.$$

Determinant:

$$\Delta = 4\alpha^2 s^2 - (s + 2\alpha c)(s - 2\alpha c).$$

Cramer for B :

$$B = \frac{\dot{\alpha}(s + 2\alpha c)}{2\Delta}.$$

Cramer for D :

$$D = -\frac{2\alpha s \dot{\alpha}}{\Delta}.$$

Insert B, D into reduced f :

$$f(\theta) = \frac{\dot{\alpha}(s + 2\alpha c)}{2\Delta}(\sin 2\theta - 2\theta \cos 2\theta) - \frac{2\alpha s \dot{\alpha}}{\Delta} \theta \sin 2\theta.$$

Factor $\dot{\alpha}/(2\Delta)$:

$$f(\theta) = \frac{\dot{\alpha}}{2\Delta} [(s + 2\alpha c)(\sin 2\theta - 2\theta \cos 2\theta) - 4\alpha s \theta \sin 2\theta].$$

Rewrite with overall minus sign to match quoted form:

$$f(\theta) = -\frac{\dot{\alpha}}{2\Delta} [4\alpha s \theta \sin 2\theta - (s + 2\alpha c)(\sin 2\theta - 2\theta \cos 2\theta)].$$

Multiply by r^2 :

$$\boxed{\psi = -\frac{\dot{\alpha} r^2}{2} \frac{(4\alpha \sin 2\alpha)(\theta \sin 2\theta) - (\sin 2\alpha + 2\alpha \cos 2\alpha)(\sin 2\theta - 2\theta \cos 2\theta)}{4\alpha^2 \sin^2 2\alpha - (\sin 2\alpha + 2\alpha \cos 2\alpha)(\sin 2\alpha - 2\alpha \cos 2\alpha)}} \quad (3)$$

which matches the quoted formula with $\Omega = \dot{\alpha}$.

Breakdown scales. Local Reynolds number with $U \sim r\dot{\alpha}$:

$$\text{Re}_{\text{loc}} = \frac{Ur}{\nu} \sim \frac{r^2 \dot{\alpha}}{\nu}.$$

Stokes validity demands:

$$\frac{r^2 \dot{\alpha}}{\nu} \lesssim 1.$$

Solve for r :

$$l_1 = \sqrt{\nu/\dot{\alpha}}.$$

End-effects of finite plate length set a second scale:

$$l_2 = L.$$

$$\boxed{l_1 = \sqrt{\nu/\dot{\alpha}}, \quad l_2 = L.}$$

(d) Volume flux through cylinder of radius r

Flux per unit length:

$$Q(r) = \int_0^\alpha v_r r \, d\theta.$$

Substitute $v_r = (1/r)\partial_\theta \psi$:

$$Q(r) = \int_0^\alpha \frac{\partial \psi}{\partial \theta} \, d\theta.$$

Evaluate antiderivative:

$$Q(r) = \psi(r, \alpha) - \psi(r, 0).$$

Use $\psi = r^2 f(\theta)$ with $f(0) = 0$, $f(\alpha) = -\frac{1}{2}\dot{\alpha}$:

$$\boxed{Q(r) = -\frac{1}{2}\dot{\alpha} r^2.} \quad (4)$$

Geometric meaning: rate of area swept inside radius r as α grows is $\frac{1}{2}r^2\dot{\alpha}$, so fluid is drawn in across the cylinder at exactly that rate (mass conservation).

(e) Free hinge: pressure, force, scallop theorem

Pressure. With ψ known, $\mathbf{v}$ is known; integrate Stokes (??):

$$\nabla p = \mu \nabla^2 \mathbf{v} = -\mu \nabla \times (\omega \hat{\mathbf{z}}).$$

Line-integrate from a reference point; fix the constant by matching ambient pressure at $r = L$.

Force scaling. Stokes flow is linear in boundary velocity, so $p \sim \mu \dot{\alpha}$ (independent of r to leading order); plate length L :

$$F \sim p \cdot L.$$

$$F \propto \mu L \dot{\alpha}.$$

Scallop theorem. Periodic $\alpha(t)$, period T . Net impulse:

$$\mathcal{I} = \oint F dt.$$

Substitute $F \propto \mu L \dot{\alpha}$:

$$\mathcal{I} \propto \mu L \oint \dot{\alpha} dt.$$

Integrate $\dot{\alpha}$:

$$\mathcal{I} \propto \mu L [\alpha(T) - \alpha(0)].$$

Apply periodicity $\alpha(T) = \alpha(0)$:

$$\mathcal{I} = 0.$$

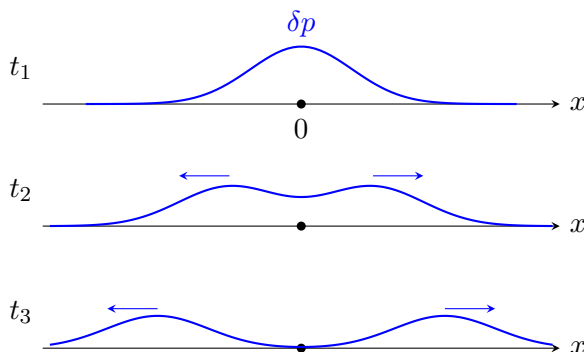
A reciprocal one-DOF gait produces no net displacement (Purcell 1977). Cyclic opening and closing of a hinged scallop cannot generate propulsion at low Re; non-reciprocal gaits (e.g. rotating helical flagella) are required.

Question 2 — Sound waves in a moving medium

Problem. Mechanism of sound; convected wave equation in a uniform stream $U_0 \hat{\mathbf{x}}$; dispersion; observers up- and downstream; supersonic and transonic cases.

(a) Mechanism of sound propagation

A local compression raises p above ambient. Pressure gradient $-\partial_x p$ accelerates neighbouring fluid outward (Newton: $\rho_0 \partial_t v = -\partial_x p$). Mass piles into next layer; that layer compresses; the bump translates. Speed: inertia/compressibility balance $c_0 = \sqrt{\partial p / \partial \rho|_{\rho_0}}$. Pulse splits into left- and right-going branches (d'Alembert).



(b) Linearisation: convected wave equation

Write $\rho = \rho_0 + \rho_1$, $p = p_0 + p_1$, $v = U_0 + v_1$.

Continuity:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0.$$

Substitute $\rho = \rho_0 + \rho_1$, $v = U_0 + v_1$:

$$\frac{\partial \rho_1}{\partial t} + \frac{\partial}{\partial x} [(\rho_0 + \rho_1)(U_0 + v_1)] = 0.$$

Expand the product:

$$\frac{\partial \rho_1}{\partial t} + \frac{\partial}{\partial x} [\rho_0 U_0 + \rho_0 v_1 + U_0 \rho_1 + \rho_1 v_1] = 0.$$

Drop the constant $\rho_0 U_0$ and the quadratic $\rho_1 v_1$:

$$D\rho_1 + \rho_0 \frac{\partial v_1}{\partial x} = 0, \quad D \equiv \frac{\partial}{\partial t} + U_0 \frac{\partial}{\partial x}. \quad (5)$$

1-D inviscid Euler:

$$\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = 0.$$

Substitute background plus perturbation:

$$\frac{\partial v_1}{\partial t} + (U_0 + v_1) \frac{\partial v_1}{\partial x} + \frac{1}{\rho_0 + \rho_1} \frac{\partial p_1}{\partial x} = 0.$$

Drop quadratic $v_1 \partial_x v_1$ and expand $1/(\rho_0 + \rho_1) \approx 1/\rho_0$:

$$Dv_1 = -\frac{1}{\rho_0} \frac{\partial p_1}{\partial x}. \quad (6)$$

Polytropic state $p = p_0(\rho/\rho_0)^\gamma$; differentiate at $\rho = \rho_0$:

$$\left. \frac{dp}{d\rho} \right|_{\rho_0} = \gamma p_0/\rho_0 \equiv c_0^2.$$

Linearise:

$$p_1 = c_0^2 \rho_1. \quad (7)$$

Apply D to (??):

$$D^2 \rho_1 + \rho_0 D \left(\frac{\partial v_1}{\partial x} \right) = 0.$$

Commute D and ∂_x :

$$D^2 \rho_1 + \rho_0 \frac{\partial (Dv_1)}{\partial x} = 0.$$

Substitute (??):

$$D^2 \rho_1 + \frac{\partial}{\partial x} \left(-\frac{\partial p_1}{\partial x} \right) = 0.$$

Simplify:

$$D^2 \rho_1 = \frac{\partial^2 p_1}{\partial x^2}.$$

Substitute (??):

$$D^2 \rho_1 = c_0^2 \frac{\partial^2 \rho_1}{\partial x^2}.$$

Hence

$$\boxed{\left(\frac{\partial}{\partial t} + U_0 \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} + U_0 \frac{\partial}{\partial x} \right) \rho_1 = c_0^2 \frac{\partial^2 \rho_1}{\partial x^2}.} \quad (8)$$

(c) Dispersion and snapshot at $U_0 = 0$

Plane wave $\rho_1 = \rho_0 A e^{i(kx - \omega t)}$ in (??):

$$(-i\omega + iU_0 k)^2 \rho_1 = -c_0^2 k^2 \rho_1.$$

Cancel ρ_1 :

$$(-i\omega + iU_0 k)^2 = -c_0^2 k^2.$$

Take the square root:

$$-i(\omega - U_0 k) = \pm i c_0 k.$$

Divide by $-i$:

$$\omega - U_0 k = \pm c_0 k.$$

$$\boxed{\omega = (U_0 \pm c_0)k.}$$

(9)

Invert for k at fixed ω :

$$k_{\pm} = \frac{\omega}{U_0 \pm c_0}.$$

Snapshot $U_0 = 0$, $\omega t = 2n\pi$. Outgoing-wave d'Alembert solution from source at $x = 0$:

$$\rho_1 = \rho_0 A \cos[\omega t - k_0 |x|], \quad k_0 = \omega/c_0.$$

Set $\omega t = 2n\pi$ and use even cosine:

$$\rho_1 = \rho_0 A \cos(k_0 |x|).$$

Apply (??):

$$p_1 = c_0^2 \rho_0 A \cos(k_0 |x|).$$

Take right-going wave for $x > 0$:

$$p_1 = c_0^2 \rho_0 A \cos(\omega t - k_0 x).$$

Differentiate in x :

$$\frac{\partial p_1}{\partial x} = c_0^2 \rho_0 A k_0 \sin(\omega t - k_0 x).$$

Use (??) with $U_0 = 0$:

$$\rho_0 \frac{\partial v_1}{\partial t} = -\frac{\partial p_1}{\partial x} = -c_0^2 \rho_0 A k_0 \sin(\omega t - k_0 x).$$

Integrate in t using $k_0 = \omega/c_0$:

$$v_1 = c_0 A \cos(\omega t - k_0 x).$$

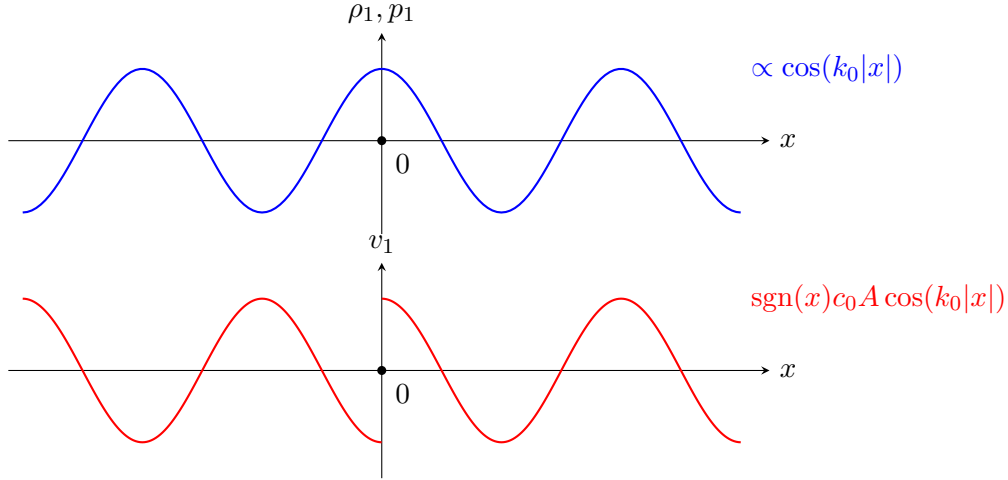
Set $\omega t = 2n\pi$:

$$v_1(x > 0) = c_0 A \cos(k_0 x).$$

Left-going wave for $x < 0$ gives the opposite sign:

$$v_1 = \text{sgn}(x) c_0 A \cos(k_0 |x|).$$

ρ_1, p_1 even in x ; v_1 odd (radiation away from source).



(d) **Two observers; supersonic;** $U_0 \rightarrow c_0$

Lab-frame wavelengths from $k_{\pm}$:

$$\lambda_{\pm} = \frac{2\pi}{|k_{\pm}|} = \frac{2\pi|U_0 \pm c_0|}{\omega}.$$

Subsonic $U_0 < c_0$. Both observers see frequency ω (steady forcing in lab frame). Downstream ($x = +L$): wavelength

$$\lambda_+ = \frac{2\pi(U_0 + c_0)}{\omega} \quad (\text{stretched}).$$

Upstream ($x = -L$): wavelength

$$|\lambda_-| = \frac{2\pi(c_0 - U_0)}{\omega} \quad (\text{compressed}).$$

Wavecounts in window of length l : $N_{\pm} = l/|\lambda_{\pm}|$:

$$\frac{N_-}{N_+} = \frac{|\lambda_+|}{|\lambda_-|} = \frac{U_0 + c_0}{c_0 - U_0}.$$

Supersonic $U_0 > c_0$. Both $k_{\pm} > 0$; both branches advected downstream. At $x = +L$ a beat pattern of two waves with $k_{\pm} = \omega/(U_0 \pm c_0)$. At $x = -L$: nothing — no characteristic reaches $x = -L$ from $x = 0$.

Limit $U_0 \rightarrow c_0^-$. $\lambda_- \rightarrow 0$, lab phase speed $c_0 - U_0 \rightarrow 0$; upstream wave is frozen and energy density diverges under finite forcing. Linear theory breaks down: nonlinear steepening (shock) and dissipation must be reintroduced — the transonic singularity.

Question 3 — Lubrication theory for a sessile drop

Problem. Thin axisymmetric drop, density ρ , viscosity μ , surface tension γ , gravity g . Justify lubrication NS, derive $p(r, z, t)$ and the parabolic $\mathbf{v}$. Volume flux gives the thin-film PDE. Steady state with $\kappa \approx -\partial_r^2 h$, contact angle θ_c , regularity at $r = 0$, volume V .

(a) Lubrication, pressure, horizontal velocity

Let H, L be vertical/horizontal scales, $\epsilon \equiv H/L \ll 1$. From continuity $U/L \sim W/H$:

$$W \sim \epsilon U \ll U.$$

Reduced Reynolds (inertia/viscous):

$$\frac{\rho U^2/L}{\mu U/H^2} = \text{Re } \epsilon^2 \ll 1.$$

Compare streamwise to vertical viscous derivatives:

$$\frac{\partial_x^2 \mathbf{v}}{\partial_z^2 \mathbf{v}} \sim \frac{H^2}{L^2} = \epsilon^2 \ll 1.$$

Drop inertia and streamwise viscous term in NS:

$$\boxed{0 \approx \mu \frac{\partial^2 \mathbf{v}}{\partial z^2} - \nabla p - \rho g \hat{\mathbf{z}}.} \quad (10)$$

Pressure. Take z -component of (??):

$$\frac{\partial p}{\partial z} = -\rho g.$$

Integrate in z :

$$p(r, z, t) = -\rho g z + f(r, t).$$

Apply Young–Laplace $p|_-^+ = p_0 - p(r, h) = -\gamma \kappa$, i.e. $p(r, h) = p_0 + \gamma \kappa$:

$$-\rho g h + f = p_0 + \gamma \kappa.$$

Solve for f :

$$f = p_0 + \rho g h + \gamma \kappa.$$

Hence

$$\boxed{p(r, z, t) = p_0 + \rho g (h - z) + \gamma \kappa(r, t).} \quad (11)$$

Horizontal velocity. Take radial component of (??):

$$\mu \frac{\partial^2 v_r}{\partial z^2} = \frac{\partial p}{\partial r}.$$

Differentiate (??) in r :

$$\frac{\partial p}{\partial r} = \rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r}.$$

Abbreviate the driving term:

$$G(r, t) \equiv \rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r}.$$

Integrate $\mu \partial_z^2 v_r = G$ once with $\partial_z v_r|_{z=h} = 0$ (no surface stress):

$$\mu \frac{\partial v_r}{\partial z} = G(z - h).$$

Integrate again with $v_r(0) = 0$ (no-slip on $z = 0$):

$$\mu v_r = G\left(\frac{1}{2}z^2 - hz\right).$$

Factor $z/2$:

$$\mu v_r = \frac{1}{2}G z(z - 2h).$$

Vector form:

$$\mathbf{v} \approx \left[\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right] \frac{z(z - 2h)}{2\mu} \hat{\mathbf{r}}. \quad (12)$$

(b) Volume flux and the thin-film equation

Flux through cylinder of radius r :

$$Q(r, t) = 2\pi r \int_0^h v_r \, dz.$$

Substitute (??):

$$Q = \frac{2\pi r}{2\mu} \left(\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right) \int_0^h z(z - 2h) \, dz.$$

Evaluate the inner integral:

$$\int_0^h z(z - 2h) \, dz = \frac{1}{3}h^3 - h^3.$$

Combine:

$$\int_0^h z(z - 2h) \, dz = -\frac{2}{3}h^3.$$

Substitute:

$$Q(r, t) = -\frac{2\pi r h^3}{3\mu} \left(\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right).$$

Annular mass balance for $r \leq r' \leq r + dr$:

$$\frac{\partial}{\partial t} (2\pi r h \, dr) = -\frac{\partial Q}{\partial r} dr.$$

Cancel dr :

$$2\pi r \frac{\partial h}{\partial t} = -\frac{\partial Q}{\partial r}.$$

Divide by $2\pi r$:

$$\frac{\partial h}{\partial t} = -\frac{1}{2\pi r} \frac{\partial Q}{\partial r}.$$

Substitute Q and rearrange:

$$\frac{\partial h}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left[r \frac{h^3}{3\mu} \left(\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right) \right]. \quad (13)$$

(c) Steady drop shape

Steady $\partial_t h = 0$ implies Q independent of r . Boundary $h(R) = 0$ forces $Q \equiv 0$. With $h > 0$ the bracket vanishes:

$$\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} = 0.$$

Integrate in r :

$$\rho g h + \gamma \kappa = K \quad (\text{const}).$$

Substitute $\kappa \approx -\partial_r^2 h$:

$$\rho g h - \gamma \frac{\partial^2 h}{\partial r^2} = K.$$

Rearrange:

$$\gamma \frac{\partial^2 h}{\partial r^2} = \rho g h - K.$$

Define capillary length:

$$l \equiv \sqrt{\gamma/(\rho g)}.$$

Divide by γ :

$$\frac{\partial^2 h}{\partial r^2} - \frac{1}{l^2} h = -\frac{K}{\gamma}.$$

General solution: homogeneous cosh, sinh plus particular $h_0 = Kl^2/\gamma$. Regularity at $r = 0$ ($\partial_r h \rightarrow 0$) kills the sinh:

$$h(r) = A_1 \cosh(r/l) + h_0, \quad h_0 \equiv Kl^2/\gamma.$$

Apply $h(R) = 0$:

$$A_1 \cosh(R/l) + h_0 = 0.$$

Solve for h_0 :

$$h_0 = -A_1 \cosh(R/l).$$

Substitute:

$$h(r) = A_1 \cosh(r/l) - A_1 \cosh(R/l).$$

Factor $-A_1$:

$$h(r) = -A_1 [\cosh(R/l) - \cosh(r/l)].$$

Set $\alpha \equiv -A_1/l > 0$:

$$h(r) = \alpha l [\cosh(R/l) - \cosh(r/l)]. \quad (14)$$

Differentiate h :

$$\frac{\partial h}{\partial r} = -\alpha \sinh(r/l).$$

Apply contact-angle condition $\partial_r h(R) = -\tan \theta_c$:

$$-\alpha \sinh(R/l) = -\tan \theta_c.$$

Solve for α :

$$\alpha = \frac{\tan \theta_c}{\sinh(R/l)}.$$

Volume:

$$V = \int_0^R 2\pi r h(r) dr.$$

Substitute (??):

$$V = 2\pi\alpha l \int_0^R [\cosh(R/l) - \cosh(r/l)] r \, dr.$$

Change variable $\xi = r/l$, $dr = l \, d\xi$, $r = l\xi$:

$$V = 2\pi\alpha l^3 \int_0^{R/l} [\cosh(R/l) - \cosh \xi] \xi \, d\xi.$$

Divide by V :

$$1 = \frac{2\pi l^3 \alpha}{V} \int_0^{R/l} [\cosh(R/l) - \cosh \xi] \xi \, d\xi. \quad (15)$$

Force balances. Bulk: ρgh (gravity hydrostatic) vs $-\gamma \partial_r^2 h$ (capillary Laplace pressure); crossover at the capillary length l . $R \ll l$: surface tension dominates, drop is a spherical cap. $R \gg l$: gravity dominates, drop is a flat puddle of thickness $\sim 2l \sin(\theta_c/2)$. Edge: Young's equation $\gamma_{SV} = \gamma_{SL} + \gamma \cos \theta_c$ fixes θ_c , hence α . Global: (??) ties R to V .

Question 4 — Complex potential and Joukowski mapping

Problem. Why simple shear has no $w(z)$. Streamfunctions of $w = \beta z^2$ and $w = \beta z^2 e^{i\alpha}$; dyed disc area. Cylinder of radius a in the second flow: find $w = Az^2 + B/z^2$, surface speed, net force. Joukowski map to ellipse: $W(Z)$ and stagnation points.

(a) **Why simple shear has no $w(z)$**

$\mathbf{v} = (\beta y, 0)$. Compute divergence:

$$\nabla \cdot \mathbf{v} = \partial_x(\beta y) + \partial_y(0) = 0.$$

Compute curl:

$$(\nabla \times \mathbf{v})_z = \partial_x v_y - \partial_y v_x = -\beta.$$

$w(z) = \phi + i\psi$ requires $\mathbf{v} = \nabla \phi$ (irrotational) and $\nabla \cdot \mathbf{v} = 0$. Simple shear has uniform vorticity $-\beta \neq 0$, so no ϕ exists, hence no $w(z)$.

(b) **Streamfunctions and the dyed disc**

(i) $w = \beta z^2$. Substitute $z = x + iy$:

$$w = \beta(x + iy)^2.$$

Expand the square:

$$w = \beta(x^2 - y^2) + 2i\beta xy.$$

Read off real and imaginary parts:

$$\phi = \beta(x^2 - y^2), \quad \psi = 2\beta xy.$$

Streamlines: $xy = \text{const}$ (rectangular hyperbolae); axes are $\psi = 0$. Pure straining flow with stagnation at origin, principal axes along x and y .

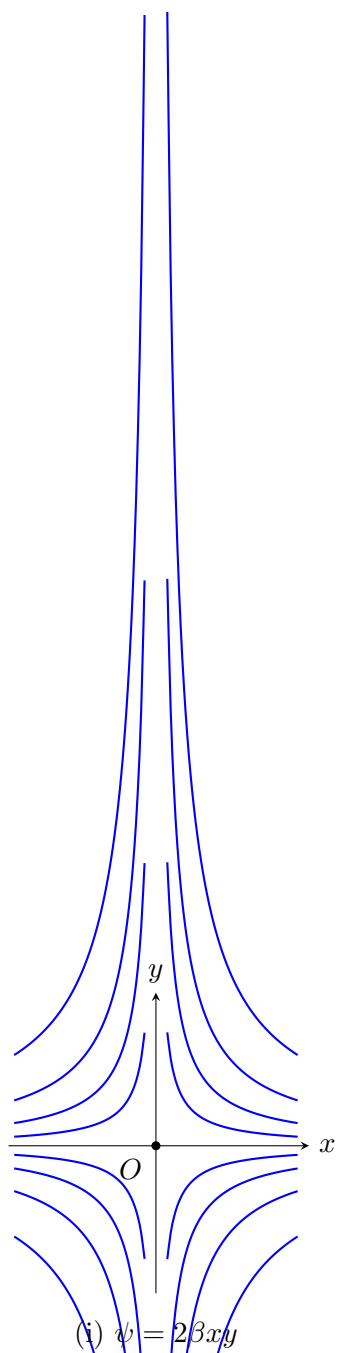
(ii) $w = \beta z^2 e^{i\alpha}$. Set $z = re^{i\theta}$:

$$w = \beta r^2 e^{2i\theta} e^{i\alpha} = \beta r^2 e^{i(2\theta + \alpha)}.$$

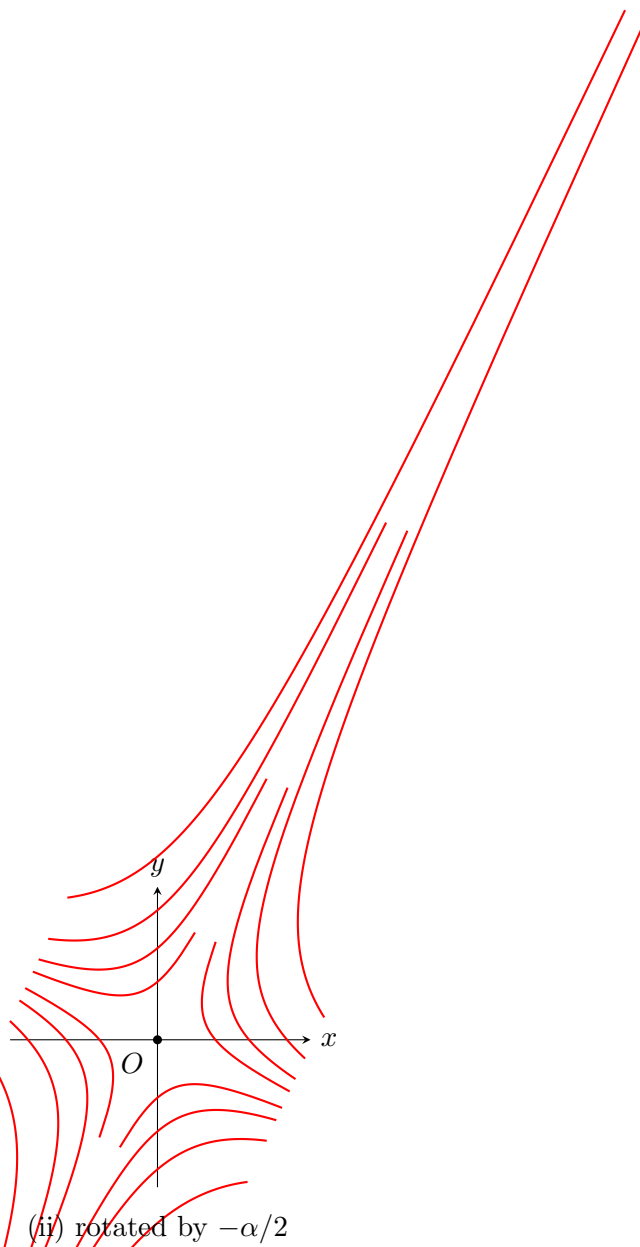
Take imaginary part:

$$\psi = \beta r^2 \sin(2\theta + \alpha).$$

Streamlines are the same hyperbolae rotated by $-\alpha/2$.



(i) $\psi = 2\beta xy$



(ii) rotated by $-\alpha/2$

Dyed disc area. 2D incompressible flow conserves material area:

$$\boxed{\text{Area}(t) = \pi R^2 \text{ for all } t.}$$

The disc deforms into an ellipse: stretches by $e^{2\beta t}$ along the extensional axis, compresses by $e^{-2\beta t}$ along the orthogonal axis. Product unity.

(c) Cylinder in the strain field

Find A, B . Far-field matching $w \sim \beta z^2 e^{i\alpha}$ fixes

$$A = \beta e^{i\alpha}.$$

Evaluate w on $z = ae^{i\theta}$:

$$w = Aa^2 e^{2i\theta} + \frac{B}{a^2} e^{-2i\theta}.$$

Impermeability requires $\text{Im } w = \text{const}$ on $|z| = a$ for all θ . Write $A = \beta(\cos \alpha + i \sin \alpha)$, $B = B_R + iB_I$. Take imaginary part:

$$\text{Im } w = \beta a^2 \sin(2\theta + \alpha) + \frac{1}{a^2} [B_I \cos 2\theta - B_R \sin 2\theta].$$

Expand $\sin(2\theta + \alpha) = \sin 2\theta \cos \alpha + \cos 2\theta \sin \alpha$:

$$\text{Im } w = \beta a^2 \sin \alpha \cos 2\theta + \beta a^2 \cos \alpha \sin 2\theta + \frac{B_I}{a^2} \cos 2\theta - \frac{B_R}{a^2} \sin 2\theta.$$

Vanish $\cos 2\theta$ coefficient:

$$\beta a^2 \sin \alpha + B_I/a^2 = 0.$$

Solve:

$$B_I = -\beta a^4 \sin \alpha.$$

Vanish $\sin 2\theta$ coefficient:

$$\beta a^2 \cos \alpha - B_R/a^2 = 0.$$

Solve:

$$B_R = \beta a^4 \cos \alpha.$$

Combine real and imaginary:

$$B = \beta a^4 (\cos \alpha - i \sin \alpha) = \beta a^4 e^{-i\alpha}.$$

Hence

$$\boxed{w(z) = \beta e^{i\alpha} z^2 + \frac{\beta a^4 e^{-i\alpha}}{z^2}.} \quad (16)$$

Surface speed. Differentiate (??):

$$\frac{dw}{dz} = 2\beta e^{i\alpha} z - \frac{2\beta a^4 e^{-i\alpha}}{z^3}.$$

Set $z = ae^{i\theta}$:

$$\frac{dw}{dz} = 2\beta a e^{i(\alpha+\theta)} - \frac{2\beta a^4 e^{-i\alpha}}{a^3 e^{3i\theta}}.$$

Simplify the second term:

$$\frac{dw}{dz} = 2\beta a e^{i(\alpha+\theta)} - 2\beta a e^{-i(\alpha+3\theta)}.$$

Factor $e^{-i\theta}$:

$$\frac{dw}{dz} = 2\beta a e^{-i\theta} \left[e^{i(\alpha+2\theta)} - e^{-i(\alpha+2\theta)} \right].$$

Apply $e^{i\phi} - e^{-i\phi} = 2i \sin \phi$:

$$\frac{dw}{dz} = 4i\beta a e^{-i\theta} \sin(2\theta + \alpha).$$

Take modulus:

$$v = \left| \frac{dw}{dz} \right| = 4a\beta |\sin(2\theta + \alpha)|. \quad (17)$$

Force on a surface element; net force. Inviscid flow: only pressure stress acts. Bernoulli (steady, irrotational) gives

$$p(\theta) = C - \frac{1}{2}\rho v^2(\theta),$$

for some constant C (set by a reference). Element force per unit length: pressure pushes inward along $-\hat{\mathbf{r}}$,

$$d\mathbf{F} = -p(\theta) \hat{\mathbf{r}} a d\theta.$$

Substitute $v^2 = 16a^2\beta^2 \sin^2(2\theta + \alpha)$:

$$d\mathbf{F} = -[C - 8\rho a^2\beta^2 \sin^2(2\theta + \alpha)] \hat{\mathbf{r}} a d\theta.$$

Integrate over θ :

$$\mathbf{F} = -aC \int_0^{2\pi} \hat{\mathbf{r}} d\theta + 8\rho a^3\beta^2 \int_0^{2\pi} \sin^2(2\theta + \alpha) \hat{\mathbf{r}} d\theta.$$

The constant piece vanishes since $\int_0^{2\pi} \hat{\mathbf{r}} d\theta = 0$:

$$\mathbf{F} = 8\rho a^3\beta^2 \int_0^{2\pi} \sin^2(2\theta + \alpha) \hat{\mathbf{r}}(\theta) d\theta.$$

Use $\sin^2(2\theta + \alpha) = \frac{1}{2}[1 - \cos(4\theta + 2\alpha)]$:

$$\mathbf{F} = 4\rho a^3\beta^2 \int_0^{2\pi} [1 - \cos(4\theta + 2\alpha)] \hat{\mathbf{r}}(\theta) d\theta.$$

Each integral vanishes: $\int \hat{\mathbf{r}} d\theta = 0$ and $\int \cos(4\theta + 2\alpha) \hat{\mathbf{r}} d\theta = 0$ since $\hat{\mathbf{r}} \propto e^{\pm i\theta}$ has no $\pm 4\theta$ Fourier content:

$$\mathbf{F} = \mathbf{0} \quad (\text{d'Alembert's paradox}). \quad (18)$$

(d) Joukowski mapping and stagnation points

Mapped potential. Invert $Z = z + c^2/z$ for z :

$$z^2 - Zz + c^2 = 0.$$

Quadratic formula, branch $z \rightarrow Z$ as $|Z| \rightarrow \infty$:

$$z(Z) = \frac{1}{2}(Z + \sqrt{Z^2 - 4c^2}).$$

Far-field $z \sim Z$ means $w(z(Z)) \sim \beta Z^2 e^{i\alpha}$ matches the required ellipse asymptote. Compose:

$$W(Z) = \beta e^{i\alpha} [z(Z)]^2 + \frac{\beta a^4 e^{-i\alpha}}{[z(Z)]^2}, \quad z(Z) = \frac{1}{2}(Z + \sqrt{Z^2 - 4c^2}). \quad (19)$$

Stagnation points. Apply chain rule:

$$\frac{dW}{dZ} = \frac{dw/dz}{dZ/dz}.$$

Compute $dZ/dz = 1 - c^2/z^2$; it vanishes at $z = \pm c$, both inside $|z| < a$ (since $c < a/2 < a$): these are outside the physical fluid domain, so map singularities only. Set $dw/dz = 0$ using (??):

$$2\beta e^{i\alpha} z - \frac{2\beta a^4 e^{-i\alpha}}{z^3} = 0.$$

Multiply by $z^3/(2\beta)$:

$$e^{i\alpha} z^4 = a^4 e^{-i\alpha}.$$

Solve for z^4 :

$$z^4 = a^4 e^{-2i\alpha}.$$

Take fourth roots:

$$z_n = a e^{i(n\pi - \alpha)/2}, \quad n = 0, 1, 2, 3.$$

These lie on $|z| = a$, on the cylinder surface. Apply Joukowski with $\theta_n \equiv (n\pi - \alpha)/2$:

$$Z_n = z_n + \frac{c^2}{z_n} = a e^{i\theta_n} + \frac{c^2}{a} e^{-i\theta_n}.$$

Split into real and imaginary parts using $e^{\pm i\theta_n} = \cos \theta_n \pm i \sin \theta_n$:

$$Z_n = \left(a + \frac{c^2}{a}\right) \cos \theta_n + i \left(a - \frac{c^2}{a}\right) \sin \theta_n.$$

Substitute $\theta_n = (n\pi - \alpha)/2$:

$$\boxed{Z_n = \left(a + \frac{c^2}{a}\right) \cos \frac{n\pi - \alpha}{2} + i \left(a - \frac{c^2}{a}\right) \sin \frac{n\pi - \alpha}{2}, \quad n = 0, 1, 2, 3.} \quad (20)$$

The four stagnation points lie on the ellipse with semi-axes $(a + c^2/a), (a - c^2/a)$. Limit $\alpha \rightarrow 0$: tips of the ellipse on its principal axes, as required by symmetry.